

Comparing LLM and human peer reviews and ratings of social science research using data from Unjournal.org evaluations

David Reinstein

Valentin Klotzbücher

2025-09-06

We will build and refine LLM tools to generate peer-reviews and ratings of impactful research, and compare these with human experts' work (esp. from Unjournal.org): to benchmark performance, understand AI's research taste, and develop tools to improve research evaluation and dissemination.

Table of contents

1	Project Overview	4
1.1	Motivation and questions	4
1.2	Related work	5
2	Numerical ratings	6
3	LLM vs. Human Ratings	7
3.1	Overview	7
3.2	Data & Harmonization	7
3.3	Distribution of ratings	7
3.4	Differences by paper and metric	8
3.5	Agreement metrics	9
	References	13

1 Project Overview

Warning

Work in progress. Pages, metrics, and comparisons are under active development. Expect rough edges and frequent updates.

1.1 Motivation and questions

Is AI good at peer-reviewing? Does it offer useful and valid feedback? Can it predict how human experts will rate research across a range of categories? How can it help academics do this “thankless” task better? Is it particularly good at spotting errors? Are there specific categories, e.g. spotting math errors or judging real-world relevance, where it does surprisingly well or poorly? How does its “research taste” compare to humans?

If AI research-evaluation works it could free up a lot of scientific resources – perhaps \$1.5 billion/year in the US alone (Aczel, Szaszi, and Holcombe (2021))) – and offer more continual and detailed review, helping improve research. It could also help characterize methodological strengths/weaknesses across papers, aiding training and research direction-setting. Furthermore, a key promise of AI is to directly improve science and research. Understanding how AI engages with research evaluations may provide a window into its values, abilities, and limitations.

In this project, we are testing the capabilities of current large language models (LLMs), illustrating whether they can generate research paper evaluations comparable to expert human reviews. The Unjournal systematically prioritizes ‘impactful’ research and pays for high-quality human evaluations, structured quantified ratings, claim identification and assessment, and predictions. In this project, we use an AI (OpenAI’s GPT-5 model for now) to review social science research papers under the same criteria used by human reviewers for The Unjournal.

Each paper is assessed on specific dimensions – for example, the strength of its evidence, rigor of methods, clarity of communication, openness/reproducibility, relevance to global priorities, and overall quality. The LLM will provide quantitative scores (with uncertainty intervals) on these criteria and produce a written evaluation

Our initial dataset will include research papers that have existing human evaluations. For each paper, the AI will generate: (1) numeric ratings on the defined criteria, (2) identification of the paper’s key claims, and (3) a detailed review discussing the paper’s contributions and weaknesses. We will then compare the AI-generated evaluations to the published human evaluations.

Next, we will focus on papers currently under evaluation, ie where no human evaluation exists yet and we can rule out any contamination.

1.2 Related work

Luo et al. (2025) survey LLM roles from idea generation to peer review, including experiment planning and automated scientific writing. They highlight opportunities (productivity, coverage of long documents) alongside governance needs (provenance, detection of LLM-generated content, standardizing tooling) and call for reliable evaluation frameworks.

Eger et al. (2025) provide a broad review of LLMs in science and a focused discussion of AI-assisted peer review. They emphasize: (i) peer-review data is scarce and concentrated in CS/OpenReview venues; (ii) targeted assistance that preserves human autonomy is preferable to end-to-end reviewing; and (iii) ethics and governance (bias, provenance, detection of AI-generated text) are first-class constraints.

Zhang and Abernethy (2025) propose deploying LLMs as quality checkers to surface critical problems instead of generating full narrative reviews. Using papers from WITHDRARXIV and an automatic evaluation framework that leverages “LLM-as-judge,” they find the best performance from top reasoning models but still recommend human oversight.

2 Numerical ratings

We evaluate each paper along the following dimensions: **claims & evidence**, **methods**, **logic & communication**, **open science**, **global relevance**, and an **overall** assessment. Ratings are interpreted as percentiles relative to serious recent work in the same area. For each metric the model returns a midpoint and a 90% credible interval to communicate uncertainty.

We iterate over `papers/*.pdf`, collect raw results and write a long CSV to `results/metrics_long.csv`.

3 LLM vs. Human Ratings

3.1 Overview

This page compares the LLM-generated ratings (`gpt-5`) from the [previous section](#) with human evaluations across [the Unjournal's criteria](#). We (i) harmonize the two sources to a common set of metrics and paper IDs, (ii) compute paired scores per *(paper, metric)*, and (iii) summarize agreement using distributional views and reliability statistics.

3.2 Data & Harmonization

- Sources. LLM ratings come from `results/metrics_long.csv` (rendered in the previous chapter). Human ratings are imported from your hand-coded spreadsheet and mapped to LLM paper IDs via `UJ_map.csv`.
- Metric alignment. Human criteria are recoded to the LLM schema (e.g., `claims` → `claims_evidence`, `adv_knowledge` → `advancing_knowledge`, etc.). If humans provided both `gp_relevance` and `real_world`, we fold these into the single LLM metric `global_relevance`.
- Uncertainty fields. Where available, we carry lower/upper bounds for both LLM and humans; these are used in optional uncertainty checks but do not affect the mid-point comparisons below.

3.3 Distribution of ratings

Figure 3.1

- (a) LLM vs Human with robust CI ellipses. Horizontal span covers the union of all human raters' CIs and point ratings; vertical span from LLM CI. Centers at midpoints. Dropdown selects metric.

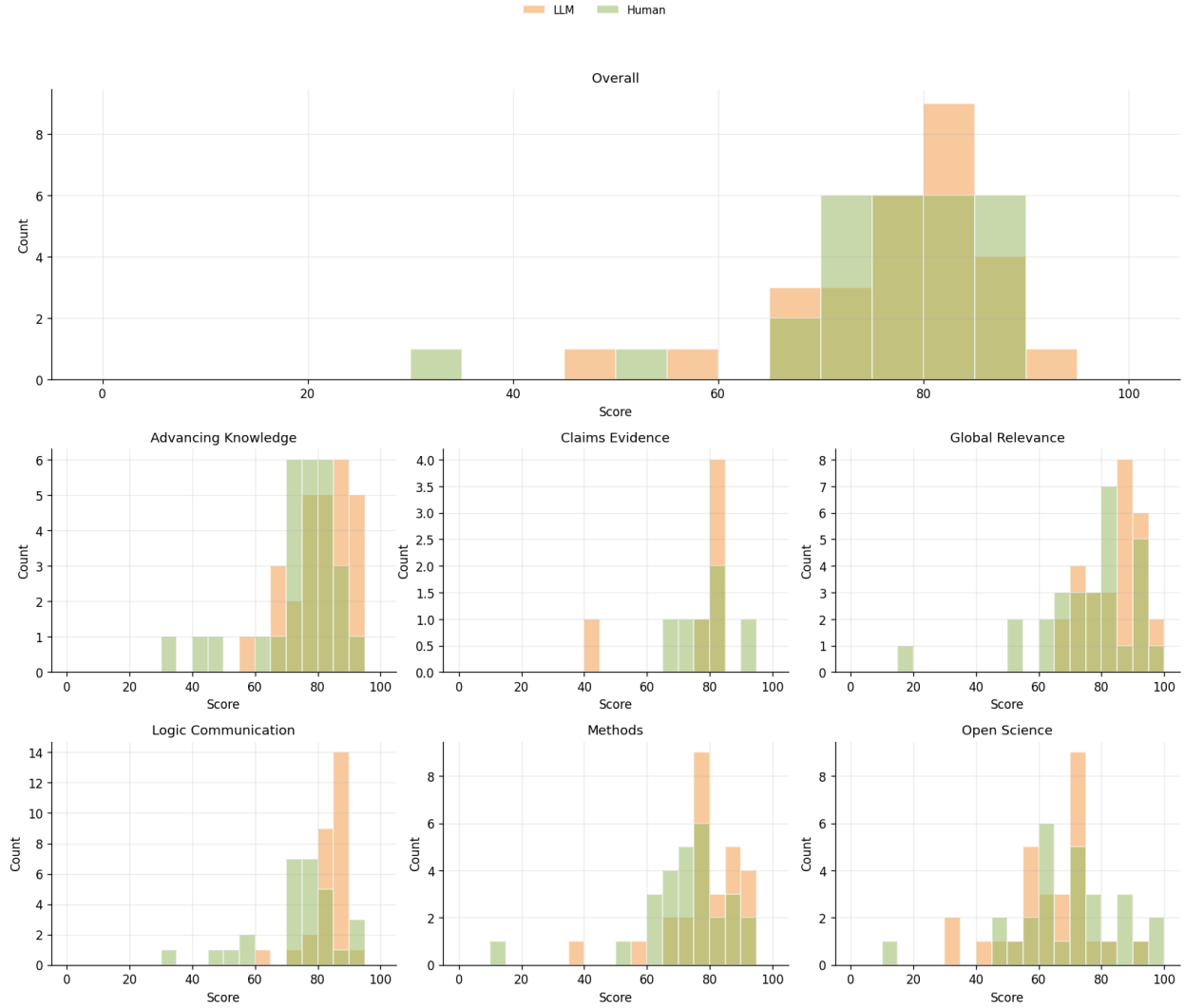
Unable to display output for mime type(s): text/html

(b)

Unable to display output for mime type(s): text/html

For each paper (selected metric), the **ellipse** is centered at the pair of midpoints (*Human*, *LLM*). The **horizontal radius** covers the *union* of all human evidence for that paper — combining individual raters’ point scores and their CIs — while the **vertical radius** reflects the LLM CI. This makes the ellipse wide enough to contain every human point horizontally (unless a point lies outside its own stated CI, which is flagged in a diagnostic check).

Figure 3.2: Distributions by metric. Top: Overall (full width). Bottom: remaining metrics in a 2×3 grid.



3.4 Differences by paper and metric

For each (*paper*, *metric*), we plot LLM — Human difference in Figure 3.3. Green means the LLM is lower than the human midpoint; orange means higher.

- Vertical stripes (by metric) suggest a metric-specific bias (LLM systematically up/down on that metric).

- Horizontal stripes (by paper) flag paper-specific disagreement (LLM consistently above/below for that paper across metrics).
- Isolated bright cells highlight individual outliers worth auditing qualitatively.

3.5 Agreement metrics

Table 3.1 summarizes agreement: Pearson (r) captures linear alignment of levels; Spearman (ρ) captures rank agreement (monotonic relationship), robust to scale differences. We report the mean and median of (LLM – Human) to summarize both average shift and central tendency of the differences and Cohen’s κ (unweighted & quadratic). Because scores are continuous, we discretize both LLM and Human using shared quantile cutpoints (5 bins) and compute the Unweighted (exact category agreement beyond chance) and the Quadratic-weighted κ that penalizes larger disagreements more than smaller ones.

As another indicator of agreement, we check whether each side’s credible interval contain the other’s point estimate, Table 3.3 shows the shares across rating categories.

Table 3.2 shows “calibration” results based on a simple linear regression; Figure 3.4 plots the width of human raters CI against the difference in ratings; and Table 3.4 lists the papers with the largest absolute difference between LLM and human raters.

Table 3.1: Agreement metrics by metric (Pearson, Spearman, mean/median bias, un-weighted/weighted).

	metric	n	pearson	spearman	mean_bias	median_bias	kappa_unwtd	kappa_quad
0	advancing_knowledge	27	0.198	0.259	7.12	6.0	0.052	0.257
1	claims_evidence	6	0.410	0.471	-3.75	-3.0	-0.154	0.400
2	global_relevance	28	0.135	0.109	8.21	8.5	-0.040	0.070
3	logic_communication	28	0.001	0.130	10.01	9.0	-0.029	0.041
4	methods	27	0.309	0.403	6.60	5.5	0.148	0.288
5	open_science	28	0.138	0.103	-6.01	-7.0	0.009	0.137
6	overall	28	0.280	0.309	1.64	1.0	0.125	0.374

Table 3.2: Calibration of LLM to Human by metric (Human = intercept + slope · LLM). Includes pre-/post- MAD.

	metric	n	slope	intercept	r2	MAD_pre	MAD_post	MAD_delta
0	advancing_knowledge	27	0.284	50.232	0.039	11.216	8.812	2.404
1	claims_evidence	6	0.220	61.234	0.168	11.250	5.859	5.391
2	global_relevance	28	0.255	53.573	0.018	14.589	12.000	2.589
3	logic_communication	28	0.002	72.823	0.000	13.464	9.802	3.663
4	methods	27	0.404	39.774	0.096	12.099	9.427	2.672
5	open_science	28	0.174	57.509	0.019	17.976	13.346	4.631
6	overall	28	0.357	47.941	0.078	9.149	7.837	1.312

Table 3.2: Calibration of LLM to Human by metric (Human = intercept + slope · LLM). Includes pre-/post- MAD.

metric	n	slope	intercept	r2	MAD_pre	MAD_post	MAD_delta
--------	---	-------	-----------	----	---------	----------	-----------

Table 3.3: CI coverage: does one interval contain the other’s point estimate?

	metric	N	human_in_llm	llm_in_human	both_cover
0	advancing_knowledge	24	0.67	0.62	0.54
1	claims_evidence	6	0.67	0.67	0.67
2	global_relevance	25	0.36	0.44	0.28
3	logic_communication	25	0.36	0.60	0.36
4	methods	24	0.62	0.54	0.50
5	open_science	25	0.48	0.40	0.28
6	overall	25	0.60	0.60	0.52

Table 3.4: Top absolute differences (LLM vs Human). Flags show whether each point lies within the other’s CI.

	paper	metric	midpoint_llm	midpoint_human	abs_diff	human_in_llm
23	Alcott et al. 2024	global_relevance	78	16.00	62.00	False
49	Bettle 2023	methods	68	10.00	58.00	False
20	Alcott et al. 2024	advancing_knowledge	90	32.50	57.50	False
63	Bruers 2021	open_science	30	80.00	50.00	False
51	Bettle 2023	logic_communication	78	30.00	48.00	False
4	Aghion et al. 2017	open_science	40	87.50	47.50	False
52	Bettle 2023	open_science	55	10.00	45.00	False
139	Kremer et al. 2022	global_relevance	95	51.25	43.75	False
82	Chuard et al. 2022	open_science	55	96.00	41.00	False
48	Bettle 2023	overall	68	32.50	35.50	False
75	Carson et al. 2023	logic_communication	85	50.00	35.00	False
39	Barberio et al. 2022	logic_communication	80	45.00	35.00	False
61	Bruers 2021	advancing_knowledge	55	90.00	35.00	False
58	Bruers 2021	overall	48	80.00	32.00	False
3	Aghion et al. 2017	logic_communication	88	57.50	30.50	False

Figure 3.3: LLM – Human difference by paper × metric (robust, centered scale; rows sorted by signed overall difference).

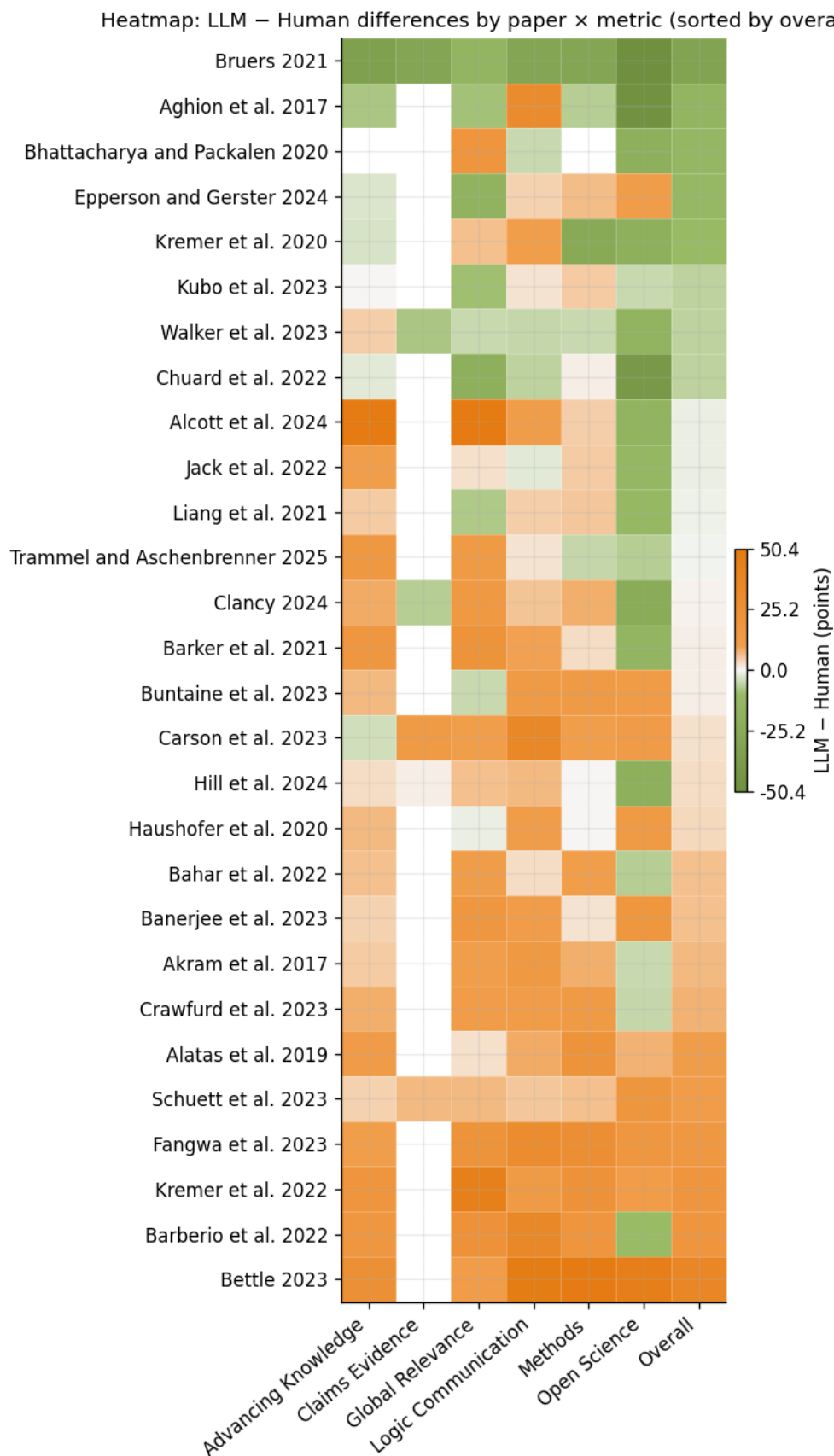
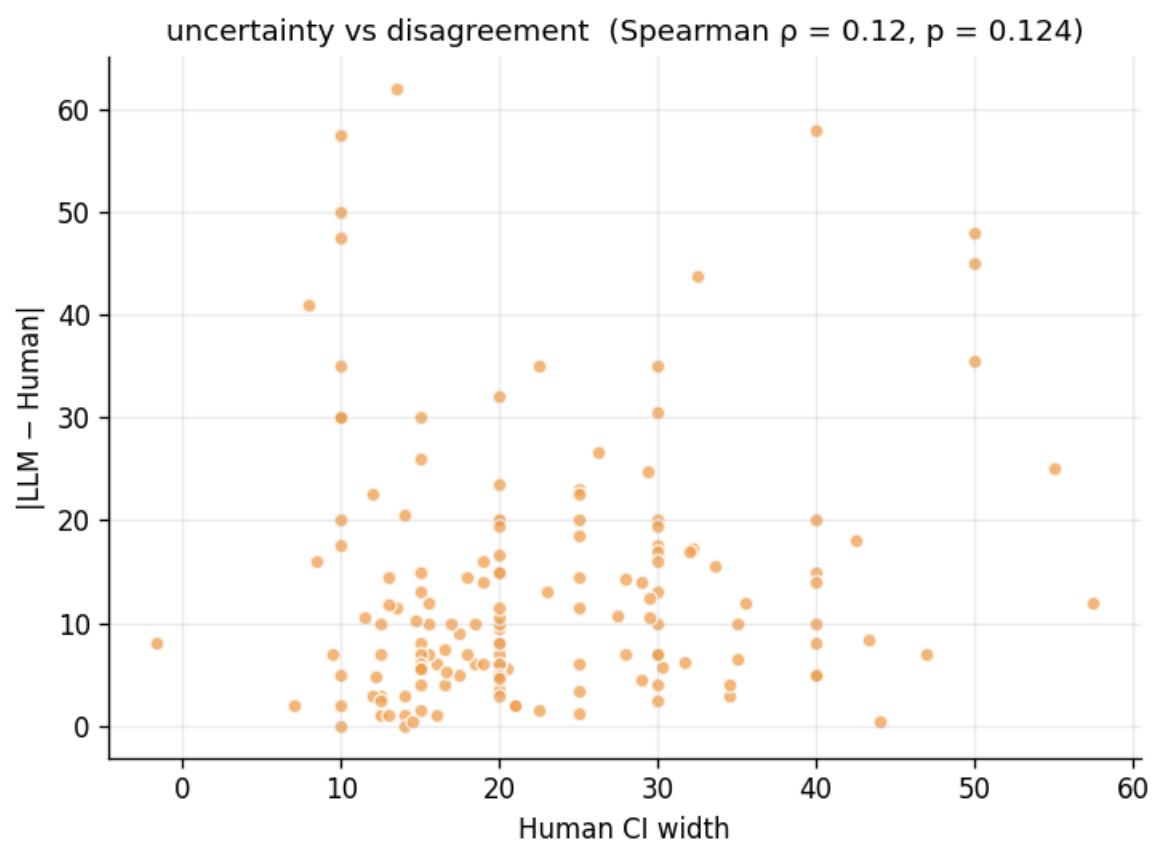


Figure 3.4: Human uncertainty (CI width) vs $|\text{LLM} - \text{Human}|$. Spearman correlation reported.



References

- Aczel, Balazs, Barnabas Szaszi, and Alex O Holcombe, “A billion-dollar donation: Estimating the cost of researchers’ time spent on peer review,” *Research integrity and peer review*, 6 (2021), 1–8 (Springer).
- Eger, Steffen, Yong Cao, Jennifer D’Souza, Andreas Geiger, Christian Greisinger, Stephanie Gross, Yufang Hou, Brigitte Krenn, Anne Lauscher, Yizhi Li, Chenghua Lin, Nafise Sadat Moosavi, Wei Zhao, and Tristan Miller, “[Transforming science with large language models: A survey on AI-assisted scientific discovery, experimentation, content generation, and evaluation](#),” *arXiv preprint arXiv:2505.05151*, (2025).
- Luo, Ziming, Zonglin Yang, Zexin Xu, Wei Yang, and Xinya Du, “[LLM4SR: A survey on large language models for scientific research](#),” *arXiv preprint arXiv:2501.04306*, (2025).
- Zhang, Tianmai M, and Neil F Abernethy, “[Reviewing scientific papers for critical problems with reasoning LLMs: Baseline approaches and automatic evaluation](#),” *arXiv preprint arXiv:2505.23824*, (2025).